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The Effect of Adjusting Screen Height and Keyboard Placement on Neck and Back Discomfort, Posture, and Muscle Activities during Laptop Work

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ABSTRACT

Laptop work may increase risks of neck and back injuries in the long term due to awkward postures and increased mechanical loads on cervical and lumbar spine. This study was intended to compare the muscle activities and postural changes as well as discomfort of the neck and back resulted from the laptop work in a conventional and adjusted setup. 20 young healthy males performed two 8-min sessions of computer-mouse work and typing in a conventional setup and in an ergonomically adjusted setup during laptop work. In the adjusted setup, the screen height was individually adjusted by a riser and the keyboard was externally connected. The posture of the cervical, thoracic, lumbar spines, and shoulder, as well as electromyographic activity of cervical and lumbar erector spinae, multifidus, upper trapezius and anterior deltoid muscles, were recorded during the first and last 2 min of each computer work session. Having completed the computer work session, furthermore, the participants scored their neck, back, and general discomfort. The very research findings implied that adjusting the screen height and keyboard placement was effective during laptop work in reducing the cervical (−4.53 deg) and thoracic flexion angles (−3.14 deg) as well as the cervical erector spinae (−10.31%) and multifidus (−15.57%) muscle activities, and resulted in lower discomfort. The typing task was associated with higher muscle activity and awkward posture compared with the mouse work. As for workstation setup, simple modifications may reduce risk factors of musculoskeletal complaints.

1. Introduction

Using laptop has become very popular owing to its portability and light weight (Osama et al., 2018; Straker et al., 1997a; Woo et al., 2016). Many people, including students, have a sedentary work pattern over long periods, especially as working with electronic devices like laptops and tablets (Ryan et al., 2011; Straker et al., 2018). Based on several studies, the continuous use of a laptop has been found out to be associated with increased risks of musculoskeletal complaints, specifically discomfort and pain in the neck, shoulder, and back regions (Allafan et al., 2018; Arshad et al., 2020; Hafeez et al., 2013; Janneck et al., 2018; Osama et al., 2018; Werth & Babski-Reeves, 2014; Woo et al., 2016). Apart from short- and long-term adverse effects caused by musculoskeletal disorders (MSD) on individual' quality of life and performance (Suka & Yoshida, 2008), MSD can result in a huge economic burden (Stewart et al., 2003). Thus, studying possible risk factors of neck and back discomfort, associated with working with a laptop, can be effective to prevent and manage such a device-related musculoskeletal complaints. As compared to desktop computers, previous studies have shown that laptop work is associated with a greater neck flexion (Reinert et al., 2018; Straker et al., 1997a; Yu et al., 2018) and discomfort (Bubric & Hedge, 2016; Straker et al., 1997a). A reason for this awkward working posture and increased discomfort can be related to the attachment of the keyboard to the screen in the design of the laptop, which does not allow for fitting such computer components to the users' anthropometrics parameters (Arshad et al., 2020; Werth & Babski-Reeves, 2014). As the laptop screen and keyboard are attached to each other, the proper screen height is purposely deemed lower than the recommended one for computer work (Asundi et al., 2012; Seghers et al., 2003). Placing the laptop on a higher workstation surface may spontaneously adjust the screen height, but this can increase an awkward posture risk in shoulder, forearm and wrist as the keyboard placement does, then, not follow the recommended ergonomic guidelines (G. G. Simoneau & Marklin, 2001; Zhu & Shin, 2012). Previous studies have indicated that an elevated keyboard height might increase shoulder abduction and flexion angles as well as shoulder and neck extensor muscle activities (Erdelyil et al., 1988; Life & Pheasant, 1984; Zhu & Shin, 2012). Particularly, increased shoulder flexion and abduction may result in increased the trapezius tonic activity (Aarås et al., 2000; Aarås & Westgaard, 1987; Chiu et al., 2015; Life & Pheasant, 1984). The elevated keyboard height can

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subsequently cause the shoulder, neck, and whole body discomfort (Balliett et al., 1996; Bergqvist et al., 1995b; Erdelyil et al., 1988; Price & Dowell, 1998; Wu & Yang, 2005; Zhu & Shin, 2012). In order to prevent such awkward forearm and shoulder postures, along with raising the level of the laptop, an external keyboard is also recommended (Arshad et al., 2020; Berkhout et al., 2004).

Several studies have been conducted to explore using laptop risers and external keyboard during laptop work. However, these studies have rarely addressed the load distribution on the neck, shoulder and back muscles objectively (Asundi et al., 2012; Berkhout et al., 2004; Sommerich et al., 2002). Asundi et al. (2012) compared postural changes while using a simple inclined stand under the laptop, as well as a commercially available riser with an external mouse and a keyboard. Their study showed that using a keyboard, riser, and a separate mouse reduced neck flexion. However, no information was observed in their findings about the level of neck, back, and shoulder muscle activities. Having simultaneously evaluated muscle activities and postural changes, further insight would be provided into the biomechanical loads associated with the workstation setup (Asundi et al., 2012; Szeto & Lee, 2002). Berkhout et al. (2004) have demonstrated that the discomfort and the mechanical load is reduced on the neck by increasing the laptop height and using an external keyboard. Additionally, the user performance (the number of correct typed entries in a unit of time) has been improved (Berkhout et al., 2004). However, this investigation has been limited to assess the discomfort and the torque around the neck, and the load distribution on the neck and back muscles has not been assessed. Low back pain has often been reported to be associated with sitting and working with computers or laptop (Anuar et al., 2016; Hafeez et al., 2013; Janneck et al., 2018; Ren et al., 2019; Stewart et al., 2003). Furthermore, the back position affects the posture and muscle load in the neck and shoulder (Aarås et al., 2000). Similarly, back muscle activities are affected by the shoulder/neck and upper back postures (Kamil & Dawal, 2015; Keyserling et al., 2005; Lee et al., 2017). Therefore, it is important to evaluate the full picture of neck, shoulder, and back posture and muscle activities during laptop work. Although computer mouse work and typing on a keyboard are reckoned very typical computer tasks, both tasks have been shown to impose quite different demands in terms of neck and shoulder posture and muscle activities (Aarås et al., 2000; Szeto et al., 2009; Won et al., 2009). In addition, previous studies reported somewhat contradictory findings as comparing mouse work and typing task in terms of difference of neck and shoulder posture and muscle activities (Cooper & Straker, 1998; J. Dennerlein et al., 2002; Szeto et al., 2009; Yang & Cho, 2012). Thus, studying the load distribution in these two tasks and the workstation setup effect on the load distribution seems mostly relevant.

As outlined above, to the best of our knowledge, no study has simultaneously investigated the workstation setup effect on the activity of neck, shoulder, and back muscles as well as the postural changes during laptop work. In particular, previous studies have not compared the workstation setup effect on back muscle activities during laptop work. Thus, this study was purposed to compare the muscle activities, postural changes, and discomfort of the neck and back during the laptop work in both conventional and adjusted setups (adjusting screen height and keyboard placement). Even though the muscle activity level is often quantified by the amplitude of EMG signals, i.e., EMG RMS, the direct measure relevance of EMG amplitude has been debated in predicting the musculoskeletal disorder risk in low load tasks (Hansson, Balogh, et al., 2000). As an alternative relative rest time (RRT) has been suggested to be effective in reflecting the muscular loading risk in such tasks, and it is less sensitive to inter-subject variability (Hansson, Nordander, et al., 2000). Previous studies have demonstrated that the low degree of RRT in neck/shoulder muscles has been being deemed a risk factor to develop work-related musculoskeletal disorders (Thorn et al., 2007; Veiersted et al., 1993). Thus, the RRT analysis along with the muscle activity level and postural change can be useful to compare laptop work in both conventional and adjusted setups. It was hypothesized that the neck, shoulder, and lumbar flexion angles, the neck and shoulder muscle activities and the neck and back discomfort would be lower (but relative rest time of the neck and shoulder muscles would be higher) during the laptop work in the adjusted setup compared with the conventional one.

2. Materials & methods

2.1. Participants and ethical consideration

Twenty healthy male university students (age: 27.05 ± 3.5 years, height: 176.9 ± 5.5 cm, weight: 71 ± 11.8 kg, BMI: 22.4 ± 3.0) participated voluntarily in this study. The participants have been frequently using computer or laptop with 5.6 ± 3.1 hours a day. The physical activity readiness questionnaire (PAR-Q) (Cardinal & Cardinal, 1995) and the modified Standardized Nordic Questionnaires (Dawson et al., 2009) were adopted to verify the participant healthiness. Histories of neurological-cardiovascular problems, significant back and neck pains (in the past 6 months) resulted in greater than 3 days off work or school, current neck, or low back pains at the time of the research were included in the study exclusion criteria. All the procedures and experimental protocols performed, including Ethics for human participants, were approved by the Institutional Review Board of Human Subjects at Kharazmi University, Tehran, Iran. The participants have been thoroughly provided with instructions so as to explain the study purpose and procedures, and they signed a consent form following highly informed.

2.2. Experimental protocol

Initially, the participants were introduced to the experimental procedure, and familiarized with the experimental computer task (as described below). Then, the skin surface preparation (i.e., shaving and rinsing with alcohol) was carried out to put the EMG electrodes according to the SENIAM guidelines (Hermens et al., 2000). Having placed the electrode, inertial motion sensors were located based on the manufacturer's recommendations (Noraxon U.S.A. Inc, 2014) to assess

posture. Finally, the workstation was set up. The tasks file and the word files related to performing the typing and mouse work tasks were prepared on the desktop to the participant, and the chair height was adjusted for each individual so that the feet were placed on the ground, and the knee angle would be 90°.

The participants sat on a chair without backrest and armrests to reduce motion artifact caused by physical contact on the electrode surface. They performed a computer task with two workstation setups, namely, the conventional and the adjusted ones. The laptop was placed on a desk, and the desk height was adjusted to be 5 cm above the elbow level corresponding to the requirements supposed for light precision tasks (Sanders & McCormick, 1998; Simoneau & Marklin, 2001). In the conventional setup, the laptop was placed so that its keyboard (spacebar key) was some 10 cm away from the near edge of the desk (Kroemer & Grandjean, 1997). In the adjusted setup, the screen height was specifically adjusted to each individual and the keyboard was externally connected. A commercially available height adjustable riser (Cooling Pad, 638 (A), china; a maximum of 18 cm in five steps) was placed underneath the laptop, and the screen height was adjusted so that the top edge of the laptop screen was at the level of the participants' eye. The distance deemed between the riser and the table edge was some 22 cm. An external keyboard (XP -product, XP-8004, China; keyboard dimension: 38.7 × 12.2 × 2.3 cm) was located 10 cm away from the table edge (to get the same distance between keyboard and the table edge in both setups), and was used by the individual in the adjusted setup (Figure 1. (A,B)).

Prior to the laptop work, EMG was recorded from the muscles (as mentioned in the next section) over an instructed 10-s period rest (a reference-sitting posture) where participants sat in a relaxed position with their hands resting on the laps. Each muscle activity was used during the reference sitting in order to calculate a resting threshold to be used in computing the relative rest time of each muscle (Samani et al., 2009a).

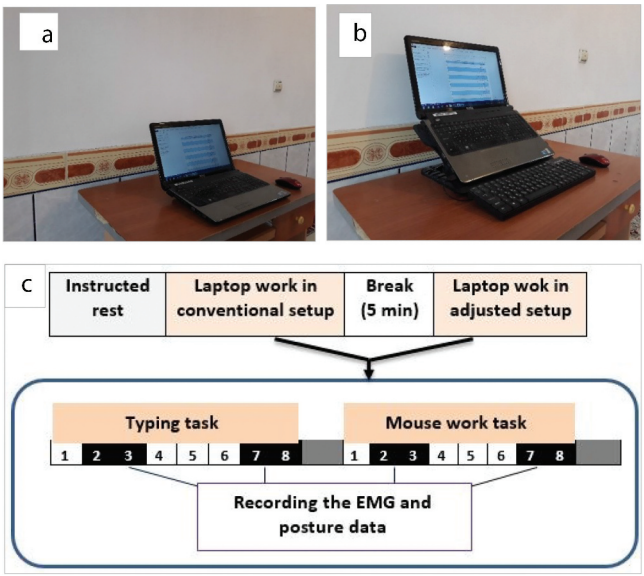


Figure 1. A & B. Laptop workstation setups. A: The conventional setup (laptop work while the laptop was on the desk). B: The adjusted setup (laptop work while use the riser and the external keyboard on the desk). C: A block diagram of the experimental procedures and data recording periods.The electromyography and posture data were recorded (black color) for 120 second at the beginning (2–3 minutes) and at the end of each task (7–8 minutes). Participants scored body discomfort after completing of each task (gray color). Further, The EMG data was recorded during instructed rest.

As for in each setup (i.e., conventional and adjusted), the computer tasks contained two 8-min segments with an 1–2 min in-between break to score discomfort and save EMG and posture data in a computer. One of the segments involved computer mouse work, and the other included typing on the laptop (Figure 1. C). In the typing task, the participant typed a text about body discomfort and sitting posture (McGill et al., 2003; Messing et al., 2008; Woo et al., 2016) displayed in the upper part of the screen (as a pdf file on the desktop) for 8 minutes with a self-selected pace. Throughout the typing task, the participants were not allowed to use the mouse or to return the written words to correct them (Xie et al., 2015). As with the mouse work tasks, the participants copied a certain section of the text (about body discomfort and sitting posture) contained in the word file on the desktop screen, and pasted into a word table already prepared for this task. The text piece to copy and paste was highlighted green, and numbered to indicate the pasting location in the table. In each cell of the table, they received a textual command to change the font size, weight, and color or highlight the text. The participants were required to make changes only with the mouse. They placed their inactive arm on the desk, and were not allowed to lean on the inactive arm during the mouse work tasks. As for both tasks, no limitation was imposed on trunk movements and leg splay but the participants were not allowed to cross their legs or change their sitting position on the seat pan. The orders of performing the mouse work and the typing task in each setup, and of adjusting the setups were randomly chosen and counter-balanced across the participants. There were five-minute breaks between the two setups.

2.3. Data recordings

The unilateral electromyographic (EMG) activities of cervical erector spinae (CES), anterior deltoid (AD), upper trapezius (UT), lumbar erector spinae (LES) and multifidus (MF) muscles as well as angular displacements of cervical, thoracic and lumbar spines and shoulder joint in sagittal plane were recorded in the first and last 2 minutes of each segment but the participants were not informed of the recording initiation. The recording operations were initiated following the first minute to provide the participants with a minute to be acclimated. Having immediately completed each task, the participants scored their level of neck, back, and general discomfort on a 0–10 scale. Discomfort was rated using an adapted body map and discomfort scale with end-point anchors of 'no discomfort: 0' and 'unbearable discomfort: 10' (Antle et al., 2015; Messing et al., 2008). Having completed the laptop work in both setups, furthermore, the participants were asked to indicate which the workstation setup was generally more comfortable (as participants' setup preference to

use laptop) and which of these setups was more comfortable for their back and neck.

A wireless surface EMG Noraxon DTS system (Noraxon Inc., USA) was used to collect electromyographic activities of CES, AD, UT, LES and MF muscles. Self-adhesive, pregelled, surface disposable Ag/AgCl electrodes (F-RG1, Skintact, Austria) were used for the signal collection. Inter electrode distance was set to 2 cm. Electrodes were then connected to wireless EMG sensors (DTS EMG sensors; Noraxon) by EMG pinch lead wires (CMRR > 100 dB, Gain: 500). The signals were transmitted to a 16-bit analogue to digital (A/D) converter receiver (Noraxon DTS receiver), and saved to a computer at a rate of 1500 Hz using the MR3 software (Version 3.10, Noraxon). The same investigator placed the electrodes on all the participants. The placement over the CES was approximately 1 cm to the right side of the C5 vertebra. Specifically, the electrodes were placed for UT at the midpoint between C7 and the acromion. As for AD, the electrodes were located at 2 cm distal and anterior to the acromion, and as for LES, the electrodes were placed at 4 cm lateral from the L1 spinous process. As with MF, the electrodes were aligned with a line from posterior superior iliac spine to the interspace between L1 and L2, and they were placed at the L5 spinous process level (i.e. about 2– 3 cm from the midline) (Basmajian & Blumenstein, 1980; Hermens & Freriks, 2017).

The angular displacement was captured by Noraxon's 3D MyoMotion analysis system consisting of a set of (1 to 16) sensors using Inertial Sensor technology (MyoMotion Clinical, Noraxon U.S.A. Inc.). In the present study, five motion sensors were located on the bony landmarks of the external occipital protuberance, C7, T12, the bony area of the sacrum, and distal section of upper arm according to the manufacturer's recommendations to measure cervical, thoracic, lumbar, and shoulder angular displacements. The procedure details of angular displacement measurement have been described (Noraxon U.S.A. Inc, 2014; Xie et al., 2018). Each sensor could provide a quaternion indicating its own spatial orientation. As a further matter, Noraxon's MR3 Software automatically converts the quaternion data into angular displacement using a rigid body model with the joint segments having a sensor attachment. The angular displacement corresponded to shoulder, cervical, thoracic and lumbar flexion/extension was measured. Prior to the experiment start, a resting trial was recorded as the participant standing upright and looking straight ahead to capture reference (null) positions of the sensors.

2.4. Data analysis

EMG signals were digitally band-pass filtered (Butterworth, 4th order, 20–500 Hz). Furthermore, a second order Butterworth digital filter of 30 Hz (high pass) was applied to remove the cardiac signal (Montes et al., 2017). To avoid any transient effect, the first and last 10 seconds of mouse work and typing tasks were removed before processing. Thus, a 100-second EMG recording was used to analyze the discussed tasks.

Root mean square (RMS) was computed across 500-ms non-overlapping epochs on EMG signals from each muscle, and then the RMS values were averaged across the epochs (St-Onge et al., 2017). As a further matter, relative rest time (RRT) was computed across 10-second non-overlapping epochs, and averaged across the epochs. RRT is reckoned a value between 0 and 100% representing the percent time that the RMS of the EMG computed over a 100 ms was below a set threshold for a minimum of 250 ms (Hermens & Vollenbroek-Hutten, 2004; Samani et al., 2009a). The resting threshold was obtained based on signals recorded during the reference sitting, and adjusted with increase or decrease in 5% steps such that RRT was at least 99% in the rest period (St-Onge et al., 2017). The mean (average) cervical, thoracic, lumbar, and shoulder flexion/extension angles were calculated across 100 seconds of the posture recording at the beginning (2–3 minutes) and at the end of each task (7–8 minutes) to assess the postural configuration during the tasks (Straker et al., 1997a; Yang & Cho, 2012; Yu et al., 2018).

2.5. Statistical analysis

SPSS version 23.0 (IBM, USA) was used to conduct the statistical analysis. A linear-mixed model (LMM) was applied to fit the outcome measures, i.e., RMS, RRT, posture and discomfort score (West et al., 2014). To find the most parsimonious and the best fit model, a "top-down" modeling strategy (West et al., 2014) was applied. Moreover, workstation setup (conventional and adjusted), tasks (mouse work and typing) and time (beginning and end) for RMS, RRT and average postural angles as well as workstation setup (conventional and adjusted) and tasks (mouse work and typing) for the discomfort rate were introduced as within-subject factors of the LMM. Having found out the fitting model, the residual of the model was checked to verify its normality. When a significant interactive effect was indicated, a Bonferroni adjustment was performed to compare the stratified dataset. In all the tests, significance was assumed at $P < .05$. In order to better compare the significant interaction results of setup × task, the percentage difference was calculated in RMS/RRT/postural angles of the same task (mouse work or typing) between conventional and adjusted setups according to the following formula:

$$[(\text{RMS}_{\text{of type}} \text{ in adjusted setup} - \text{RMS}_{\text{of type}} \text{ in conventional setup}) / \text{RMS}_{\text{of type}} \text{ in conventional setup}] \times 100.$$

3. Results

3.1. Muscle activities

The workstation setup had a significant effect on the RMS of CES and MF muscles (Table 1). In the adjusted setup, the RMS of CES and MF was significantly lower as compared to the conventional one, but the LES indicated a tendency toward a slight higher RMS in the adjusted setup (Table 1 and Figure 2). Additionally, the task type significantly affected the RMS of all muscles (Table 1). RMS of all muscles was deemed significantly higher in the typing task as compared

Table 1. The results of the statistical analysis performed to analyze the RMS and RRT of muscles during laptop work.

	FACTOR	CES	UT	AD	LES	MF
RMS	**setup**	$F_{1,\ 54} = 8.28$; $P = .01$	$F_{1,\ 55.69} = 1.39$; $P = .24$	$F_{1,\ 17.99} = .63$; $P = .44$	$F_{1,\ 78.34} = 3.70$; $P = .06$	F1, 70.36 = 5.61; $P = .02$
	task	$F_{1,\ 54} = 90.25$; $P < .01$	$F_{1,\ 55.69} = 62.01$; $P < .01$	$F_{1,\ 19.05} = 15.98$; $P < .01$	$F_{1,\ 25.03} = 52.0$; $P < .01$	F1, 26.52 = 62.05; $P < .01$
	time	$F_{1,\ 54} = .01$; $P = .97$	$F_{1,\ 55.69} = .01$; $P = .97$	$F_{1,\ 16.87} = 1.64$; $P = .22$	$F_{1,\ 67.05} = 3.69$; $P = .06$	F1, 94.79 = .12; $P = .73$
	setup × task	$F_{1,\ 72} = 2.53$; $P = .12$	$F_{1,\ 58.63} = 33$; $P = .57$	$F_{1,\ 40.81} = 5.36$; $P = .03$	$F_{1,\ 91.46} = .09$; $P = .76$	F1, 107.94 = 1.11; $P = .29$
	setup × time	$F_{1,\ 72} = 1.24$; $P = .27$	$F_{1,\ 58.63} = .15$; $P = .70$	$F_{1,\ 77.32} = .18$; $P = .67$	$F_{1,\ 56.50} = 3.80$; $P = .06$	F1, 109.31 = .74; $P = .39$
	task × time	$F_{1,\ 72} = .04$; $P = .85$	$F_{1,\ 58.63} = .01$; $P = .92$	$F_{1,\ 61.08} = 3.95$; $P = .05$	$F_{1,\ 91.29} = 1.44$; $P = .23$	F1, 107.86 = .17; $P = .68$
	Setup × task × time	$F_{1,\ 72} = 1.86$; $P = .18$	$F_{1,\ 58.63} = .95$; $P = .33$	$F_{1,\ 72.00} = .65$; $P = .42$	$F_{1,\ 91.14} = .46$; $P = .50$	F1, 94.51 = 2.84; $P = .09$
RRT	**setup**	$F_{1,\ 26.67} = 8.61$; $P = .01$	$F_{1,\ 57} = .67$; $P = .42$	$F_{1,\ 57} = .01$; $P = .93$	$F_{1,\ 19} = .93$; $P = .35$	F1, 133 = .65; $P = .42$
	task	$F_{1,\ 18.88} = 24.91$; $P < .01$	$F_{1,\ 57} = 34.13$; $P < .01$	$F_{1,\ 57} = 30.72$; $P = .01$	$F_{1,\ 19} = 25.02$; $P < .01$	F1, 133 = 91.46; $P < .01$
	time	$F_{1,\ 32.18} = .12$; $P = .73$	$F_{1,\ 57} = .01$; $P = .90$	$F_{1,\ 57} = .86$; $P = .36$	$F_{1,\ 19} = .83$; $P = .37$	F1, 133 = .79; $P = .37$
	setup × task	$F_{1,\ 36.52} = 7.92$; $P = .01$	$F_{1,\ 76} = .01$; $P = .91$	$F_{1,\ 76} = 6.23$; $P = .01$	$F_{1,\ 76} = 3.54$; $P = .06$	F1, 133 = .12; $P = .73$
	setup × time	$F_{1,\ 36.52} = .06$; $P = .80$	$F_{1,\ 76} = .03$; $P = .86$	$F_{1,\ 76} = .33$; $P = .57$	$F_{1,\ 76} = .37$; $P = .54$	F1, 133 = .60; $P = .44$
	task × time	$F_{1,\ 36.52} = .07$; $P = .80$	$F_{1,\ 76} = .06$; $P = .80$	$F_{1,\ 76} = 1.22$; $P = .27$	$F_{1,\ 76} = 2.22$; $P = .14$	F1, 133 = .05; $P = .81$
	Setup × task × time	$F_{1,\ 36.52} = .03$; $P = .85$	$F_{1,\ 76} = .40$; $P = .53$	$F_{1,\ 76} = .00$; $P = .99$	$F_{1,\ 76} = .42$; $P = .52$	F1, 133 = 2.92; $P = .09$

Note. RMS: root mean squared, RRT: relative resting time, CES: cervical erector spinae, UT: upper trapezius, AD: anterior deltoid, LES: lumbar erector spinae, MF: multifidus.

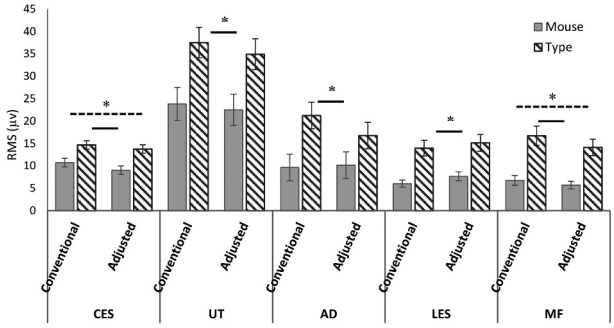


Figure 2. The mean and standard error of root mean square (RMS) of muscles during laptop work.

CES: cervical erector spinae, UT: upper trapezius, AD: anterior deltoid, LES: lumbar erector spinae, MF: multifidus. * = Significant differences between workstation setups (-*-) and tasks (—*—), $p < .05$.

with the mouse work task (Figure 2). Interaction of setup × task had a significant effect on the AD (Table 1). RMS of the AD was much less in adjusted setup than the conventional setup during typing [−21.11%]. However, it did not have any marked sign of a difference [5.02%] in the mouse work task (Figure 2).

RRT was significantly higher for the CES in the adjusted setup compared with the conventional one (Table 1 and Figure 3). As for all muscles, RRT was significantly reckoned lower in the type task as compared with the mouse work one (Table 1 and Figure 3).

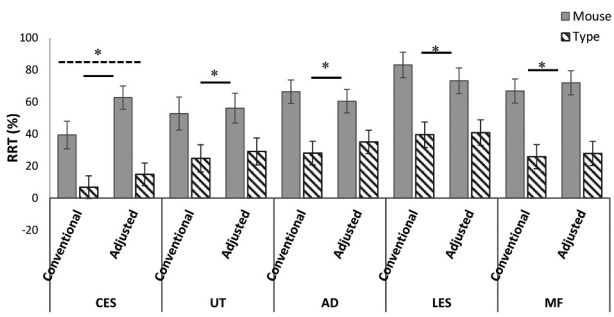


Figure 3. The mean and standard error of relative rest time (RRT) of muscles during laptop work.

CES: cervical erector spinae, UT: upper trapezius, AD: anterior deltoid, LES: lumbar erector spinae, MF: multifidus. * = Significant differences between workstation setups (–*–) and tasks (–*), $p < .05$.

Interaction of setup × task indicated a significant effect on RRT of the CES and AD muscles (Table 1). RRT was reckoned much more in the CES during typing in adjusted setup as compared with the conventional [118.84%], but this difference was modest in the mouse work task [59.26%] (Figure 3). Finally, RRT was reckoned much more for the AD in the adjusted setup compared with the conventional setup [24.68%] during typing. However, this difference was insignificant in the mouse work task [−8.87%] (Figure 3).

3.2. Posture

The workstation setup indicated a significant effect on the cervical and thoracic flexion angles (Table 2). Furthermore, the task type was demonstrated to have a significant effect on the shoulder, neck, thoracic and lumbar flexion angles (Table 2). The absolute posture difference between workstation setups (conventional-adjusted) and tasks (typing-mouse work) has been shown in Figure 4. As with the adjusted setup, the cervical and thoracic segments of the spine were positioned in a less flexed posture compared with the conventional setup (Table 2 and Figure 4, A). The cervical, thoracic, and shoulder flexion was significantly reckoned more in the type task as compared with the mouse work one. However, the lumbar flexion was less in the type task (Table 2 and Figure 4, B). Interaction of setup × task had a significant effect on the cervical, thoracic, and shoulder flexion (Table 2). The cervical and thoracic flexion was reckoned much more in the conventional setup than the adjusted setup [6.80 and 4.75 deg, respectively] during mouse work task. However, this very difference was modest in the typing task [2.27 and 1.52 deg, respectively]. During typing, the shoulder flexion was much more in the conventional setup as compared with the adjusted one [4.66 deg] whereas it did not have any marked sign of a difference [0.52 deg] in the mouse work task.

3.3. Discomfort

Laptop work has increased discomfort in the neck and back in both workstation setups. Nonetheless, the neck discomfort tended to be lower in the adjusted setup compared with the conventional setup (Table 3 and Figure 5). Further, 84% of the participants reported that the adjusted setup was generally more comfortable, and they preferred this setup for work

Table 2. The results of the statistical analysis performed to analyze the mean flexion angles of cervical, thoracic, lumbar spine and shoulder during laptop work.

FACTOR	CERVICAL	THORACIC	LUMBAR	SHOULDER
setup	$F_{1,\ 54.44} = 11.43$; $P < .01$	$F_{1,\ 17} = 16.04$; $P < .01$	$F_{1,\ 56.86} = .01$; $P = .91$	$F_{1,\ 51.52} = 2.05$; $P = .16$
task	$F_{1,\ 54.44} = 36.09$; $P < .01$	$F_{1,\ 17} = 24.79$; $P < .01$	$F_{1,\ 56.86} = 21.44$; $P < .01$	$F_{1,\ 51.42} = 6.57$; $P = .01$
time	$F_{1,\ 54.19} = .08$; $P = .77$	$F_{1,\ 17} = .01$; $P = .88$	$F_{1,\ 56.73} = .38$; $P = .54$	$F_{1,\ 51.42} = .33$; $P = .57$
setup × task	$F_{1,\ 71.70} = 20.92$; $P < .01$	$F_{1,\ 68} = 36.43$; $P < .01$	$F_{1,\ 63.61} = .22$; $P = .64$	$F_{1,\ 67.94} = 9.19$; $P < .01$
setup × time	$F_{1,\ 70.51} = .28$; $P = .60$	$F_{1,\ 68} = 4.18$; $P = .04$	$F_{1,\ 62.56} = .36$; $P = .55$	$F_{1,\ 66.61} = 2.09$; $P = .15$
task × time	$F_{1,\ 70.51} = .01$; $P = .95$	$F_{1,\ 68} = .76$; $P = .39$	$F_{1,\ 62.56} = .11$; $P = .74$	$F_{1,\ 66.61} = 2.84$; $P = .10$
Setup × task × time	$F_{1,\ 70.51} = .74$; $P = .39$	$F_{1,\ 68} = .14$; $P = .71$	$F_{1,\ 62.56} = .70$; $P = .40$	$F_{1,\ 66.61} = .18$; $P = .68$

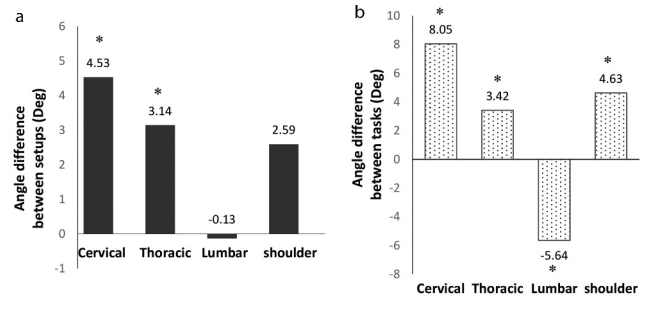


Figure 4. The mean of the differences in posture (flexion/extension angles) of cervical, thoracic, lumbar spines and shoulder between workstation setups (A: conventional setup- adjusted setup) and tasks (B: type task- mouse task) during laptop work.

* = Significant differences between workstation setups and tasks, $p < .05$. Positive differences indicates higher flexion angles in the conventional setup/type task whereas negative differences indicates higher flexion angles in the adjusted setup/mouse task.

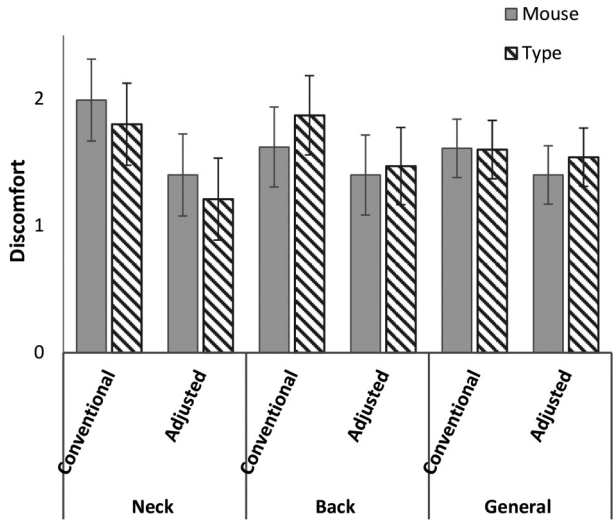


Figure 5. The mean and standard error of discomfort during laptop work.

with laptops. Also, some 67% of them reported that it was specifically less discomforting for their neck and back.

4. Discussion

As compared with the conventional setup, the results indicated that the activity level of CES and MF muscles as well as the cervical and thoracic flexion angles were lower (and CES resting was higher) during the laptop work in the adjusted setup, but its LES indicated a slightly higher activity level. As for all the muscles, the activity level as well as the cervical, thoracic, and shoulder flexion was reckoned higher (and RRT of all muscles was deemed lower). However, the lumbar flexion was lower in the typing task as compared with the mouse work. In the adjusted setup, neck discomfort tended to be lower compared with the conventional setup, and adjusted setup was generally less discomforting to participants' neck and back.

For supporting the hypothesis and according to Villanueva et al. (1997) findings, the higher activity level (and lower resting) of CES is concomitant with the higher neck flexion angle in the conventional setup as compared with the adjusted setup. Villanueva et al. (1997) examined the influence of adjusting screen height in video display terminals. They found lower neck and shoulder muscle activities with increased screen height during mouse work. They have also reported a high correlation coefficient between the extent of neck flexion and the neck extensor muscle activities. Further, Yu et al. (2018) have shown that, compared to the desktop computer, using tablet, and laptop resulted in increased neck flexion and shoulder elevation. Hence, it seems that prolonged laptop work may further increase the risk of developing neck disorders in the conventional setup, as the participants experienced a more flexed neck and an increase in the mechanical load imposed on the neck (Kleine et al., 1999; Kusumawati & Muslim, 2019; Reinert et al., 2018; Sommerich et al., 2002; Szeto & Lee, 2002; Yu et al., 2018).

One of the important and interesting findings obtained as a result of the present study was that MF muscle activities were lower but LES muscle activities tended to be slightly higher in the adjusted setup as compared with the conventional one. Nevertheless, no significant difference was found out in the lumbar flexion angle between the conventional and adjusted setups. However, the thoracic flexion angle was reckoned higher in the conventional setup compared with the adjusted setup (Figure 3). Therefore, it appears that the discrepancy between the trend of MF and LES muscle activities can be attributed, across the workstation setups, to the difference of the thoracic posture in workstation setups (Aarås et al., 2000; Kamil & Dawal, 2015; Keyserling et al., 2005) and likely the functional role of these muscles. Even

Table 3. The results of the statistical analysis performed to analyze body discomfort during laptop work.

FACTOR	NECK DISCOMFORT	BACK DISCOMFORT	GENERAL DISCOMFORT
setup	$F_{1,\ 40} = 3.09$; $P = .09$	$F_{1,\ 41.96} = 1.50$; $P = .23$	$F_{1,\ 20} = .27$; $P = .61$
task	$F_{1,\ 40} = .31$; $P = .58$	$F_{1,\ 41.96} = .41$; $P = .53$	$F_{1,\ 20} = .14$; $P = .71$
setup × task	$F_{1,\ 20} = .00$; $P = 1.00$	$F_{1,\ 20.54} = .18$; $P = .67$	$F_{1,\ 20} = 1.12$; $P = .30$

though LES and MF synergistically counter-act the spinal flexion, MF has the most advantageous lever arm for posterior sagittal rotation as it mainly inserts at the spinous processes whereas LES inserts at the transverse processes. Thus, it is expected that MF plays a more active role in holding the flexed trunk (Mork & Westgaard, 2009). A study conducted by Kamil and Dawal (2015), indicated that a significant correlation was observed among the pelvic, trunk angles and back muscles activities. Also, the findings revealed that the participants demonstrated the least level of MF muscle activities when sitting with an extended upper trunk posture (Kamil & Dawal, 2015). As compared with adjusted setup, the observed longer period of postures with increased thoracic spine flexion in the conventional setup may have a negative effect on the thoracic spine musculoskeletal structures and other related structures such as shoulder joint (Briggs et al., 2007; Clark & Lucett, 2010; Kleine et al., 1999; Klussmann et al., 2008). Also, Excessive lumbar spine flexion may increase the tension in the lumbar passive structures during sitting, and it may result in back pain in the long term (Jia & Nussbaum, 2018; Mork & Westgaard, 2009; Mörl & Bradl, 2013). Thus, considering the improvement in ergonomic workplace (Janneck et al., 2018; Kaliniene et al., 2016; Klussmann et al., 2008), using appropriate interventions are required to reduce thoracic and lumbar spine flexion during laptop work. Although the adjusted setup reduced the thoracic flexion and decreased MF muscle activities, this setup has not significantly affected the lumbar spine posture. Observing a flexed lumbar spine in both workstation setups may be due to the participants' a motor habit in performing computer work, rendering workstation setup adjustment ineffective in altering the lumbar spine posture.

The activity level of all muscles was reckoned higher (and muscle resting of all muscles was lower) in the typing task as compared with the mouse work task in both setups. As expected, the higher shoulder flexion was concomitant with a higher activity of the AD muscle during the typing task compared with the mouse work task (Hermans et al., 1998). During typing task, the neck was more flexed and lumbar flexion was lower than the mouse work task. This can likely be due to the fact that the participants typed with both hands, but they did the mouse work with one hand; this may likely be because of the higher shoulder flexion during typing task. In addition, the activity of the CES, LES and MF muscles might be enhanced to stabilize the spine and counteract the torque created by hand movements (Borghuis et al., 2008; McGill et al., 2003). As for the mouse task time period, it is worth noting that holding the full arm supporting on the work surface may have further highlighted the difference between the two tasks. During the typing task compared with mouse work task, higher activity level of CES and UT muscles is consistent with the findings of Sommerich et al. (2001) and Szeto et al. (2009) studies. In the study conducted by Sommerich et al. (2001), neck flexion as well as left levator scapulae, left and right sternocleidomastoid, trapezius and neck extensor muscle activities were higher during typing compared with mouse work task (Sommerich et al., 2001). Yang and Cho (2012) have also reported higher neck and shoulder flexion and upper trapezius muscle activities during typing task compared with mouse work task (Yang & Cho, 2012). Unlike the present study, in a study conducted by Dennerlein et al. (2002), shoulder posture was deemed the most neutral for Keyboard-intensive tasks whereas the shoulder was flexed, abducted, and externally rotated during the mouse-intensive task in order to reach out and interact with the mouse. In a pilot study, Cooper and Straker (1998) have reported a higher activity of the upper trapezius and lower activity of the anterior deltoid during keyboard task compared to mouse task. That study also showed individual differences between muscle activity for mouse and keyboard task (Cooper & Straker, 1998). The discrepancies between the present study results and the abovementioned studies may be due to differences in content of task, typing style, and the location of the keyboard and mouse (J. T. Dennerlein & Johnson, 2006). Even though the neck and shoulder posture and muscle activities can be affected by the type of tasks, trunk posture, workstation setup including the position of the mouse and dimension of keyboards and other computer components, typing style, as well as anthropometric characteristics of participants affect the posture and muscle activity (Arne Aarås et al., 2000; Cooper & Straker, 1998; J. Dennerlein et al., 2002; J. T. Dennerlein & Johnson, 2006; Otto et al., 2018; Sommerich et al., 2002; Won et al., 2009; Yang & Cho, 2012). Therefore, it seems that typing task, particularly in the conventional setup, may be associated with higher physical exposure, and may increase the likelihood of associated adverse effects of physical exposure on shoulder. Thus, the attempts to redesign the tasks are deemed important to reduce the required duration of typing in computer workstation.

The significant interactive effect of setup and task implies that flexion of the cervical and thoracic spine was much higher in the conventional setup during mouse work task than the adjusted setup. However, this difference was modest in the typing task. Due to the difference reckoned in the screen height in two workstation setups, the cervical and thoracic spine was positioned in a less flexed posture during the mouse work. Nonetheless, the role of screen height may not probably be influential during typing task because the participants may have frequently kept their gazing eye on the keyboard (Kamil & Dawal, 2015; Schüldt, 1988). Therefore, it seems that adjusting the workstation ergonomically may be more effective in improving the neck and thoracic posture during mouse work than typing. The CES muscle resting (RRT of the CES) was much higher during typing in the adjusted setup than the conventional setup. But this difference was reckoned modest in the mouse work task. This was likely due to frequent flexion and extension of the neck to shift the gaze between the screen and the keyboard in the typing task whereas the gaze would likely be kept on the screen during the mouse work. As for the typing task, the observed higher neck flexion would also support this explanation. Also, the AD muscle resting was much higher during the typing task in the adjusted setup (as well as the activity level was much lower) as compared with the conventional setup. However, such the difference was modest in the mouse work task. Further, shoulder flexion angle during typing in the adjusted setup was much lower than the conventional setup, but this difference seemed modest in the mouse work task. These observations are most likely related to the need for hand movements over the entire keyboard surface. As

compared with the adjusted setup, thus, typing task may be associated with higher physical exposure in the conventional setup, and may increase the likelihood of associated adverse effects on shoulder and neck in the long run (Chiu et al., 2015; Lale & Korhan, 2015; Sommerich et al., 2002; Straker et al., 1997b).

Neck discomfort tended to be lower in the adjusted setup as compare with the conventional setup. This difference is deemed to be due to a higher CES muscle resting and less cervical and thoracic flexion in the adjusted setup compared with the conventional one (Berkhout et al., 2004; Chiou et al., 2012; Villanueva et al., 1997; Yu et al., 2018). Based on biomechanical principles, one may assume that an upright neck position will result in less mechanical load on the neck. Furthermore, a more flexed neck results in higher activity of the cervical extensor muscles in the conventional setup (Seghers et al., 2003; Villanueva et al., 1997). Berkhout et al. (2004) have shown that, during continuous laptop work, the access to adjusting screen height, distance, and the use of an external keyboard and mouse will improve user comfort and productivity, and reduce the mechanical load on the neck. Moreover, greater increase in the neck flexion, head tilt, and neck discomfort were reported during laptop work as compared with working on a desktop computer (Bubric & Hedge, 2016; Straker et al., 1997a; Yu et al., 2018). Thus, prolonged laptop use has a greater potential to lead to over-exert the neck musculature in the conventional setup.

One of the limitations faced in this study was the use of the chair without backrest. Considering that the lumbar erector spinae and multifidus muscles sensors were placed on the back and regarding the participants were not able to lean on the backrest, a chair without backrest was used in this experiment as it has also been used in previous studies (Callaghan & McGILL, 2001; Jia & Nussbaum, 2018; Waongenngarm et al., 2016). As a further matter, the external keyboard was placed in front of the laptop in the adjusted setup. Thus, the distance between the laptop screen and the edge of the table was slightly more than the conventional setup. Although this may affect the interface between the user and the laptop, it is not expected to cause a bias in reduced estimation of physical exposure in favor of the adjusted setup. In the present study, only male students were recruited, and female students were not included. This was firstly to keep the subject pool homogenous in this study, and it was secondly caused due to legal prohibition of the engagement of male researchers in physical contacts with female participants in our country. Noting that the incidence of neck and shoulder complaints are more frequent among females than males, one should be cautious about generalizing our results to a female population (Heidari et al., 2019; Won et al., 2009; Yang & Cho, 2012), and further researches are necessary to verify whether the result will be reproduced in females. Any information was not collected on the participants' history of laptop use as compared to desktop computer use, even though most of them were frequent laptop users. It is conceivable that the participants' history of computer use would affect their response to the given task, and might affect how they perceive the level of discomfort. In spite of this limitation, our study is concerned with within-subject changes across the two conditions, and is not particularly sensitive to inter-subject variability.

We opted for a fixed-size workstation setup across all the participants to only change the studied characteristic of the workstation setup (i.e., screen height), and keep the rest of variables controlled and identical across the conditions. However, it cannot be denied the advantage of tailoring the workstation setup for each individual even though it could have made the implementation more time consuming and open for further discussions about the proper choice of set limits. Another important consideration when adjusting the height of a laptop screen during laptop work in the adjusted setup was the allowance of trunk movements. The screen height was initially adjusted in an upright seating posture. But the screen height may have been placed above the level of the participants' eye due to the trunk flexion throughout the task, and it may have led to non-neutral posterior head tilt. Despite such a limitation, subtle trunk movement may also preclude postural fixity as a positive effect during the task. However, the influence extent of such subtle postural variation is difficult in long term to precisely be assessed.

Our results do not suggest that improving the workstation setup may alone alter the extent of lumbar flexion during laptop work. Thus, a cognitive and physical training protocol may be useful to change the motor habits of sitting in a specific position (Dockrell et al., 2010; Kaliniene et al., 2016).

5. Conclusion

The results of this study indicated that the use of the ergonomically adjusted setup (using a riser under the laptop and external keyboard) compared with the conventional setup, reduced the cervical, thoracic, and shoulder flexion angles as well as the activity level of cervical erector spinae and multifidus muscles. Cervical erector spinae muscle resting also increased in the adjusted setup. Further, the adjusted setup was generally less discomforting for participants' neck and back. The activity level of all muscles as well as the cervical, thoracic, and shoulder flexion was higher in the typing task compared with the mouse work task in both setups. These findings emphasize the importance of matching the ergonomic design of the laptops to match with anthropometric scales of individuals. Given the prevalence of laptops and the high rate of laptop use, it is recommended that users who use laptop in a long-term and sedentary positions should use a riser and external keyboard as using the laptop to improve the neck and back condition, and reduce the likelihood of musculoskeletal problems. In addition, these findings imply that manufacturers and industrial designers of the laptops should consider designing laptops with adjustable screen height and keyboard placement. The typing task may be associated with higher physical exposure, and it may increase the likelihood of associated adverse effects of physical exposure on shoulder and neck in the long run. Thus, it is advisable to reduce the extent of typing task in the construct of computer tasks. Although the adjusted setup reduced the thoracic flexion and decreased MF muscle activities, this setup alone did not alter the lumbar flexion. A cognitive and physical training protocol may be useful along with improving the workstation setup to change the motor habits of sitting in a specific position.

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